- The Hilbert space $\mathcal{H}$ of any isolated quantum system can be generically described as $\mathcal{H} = \bigoplus_{\alpha=1}^{K} K_\alpha$, $K_\alpha := \mathrm{span}_t\{U^t|\psi_\alpha\rangle\}$, where U^t is the unitary governing the time evolution. $\mathrm{span}_t\{U^t|\psi_\alpha\rangle\}$ denotes the subspace spanned by time-evolution of the state $|\psi_\alpha\rangle$, and K is the number of Krylov subspaces of the system.
- Hilbert space framentation depends only on the local terms of the Hamiltonian,

$$H = \sum_j J_j \hat{h}_j ,$$

where $\hat{h}_j$ is a strictly local operator in the vicinity of site j such that $\hat{h}_i$ and $\hat{h}_j$ for $i \neq j$ need not commute and J_j are arbitrary coefficients.
- We are interested in operators $\hat{O}$ that commutes with each term, i.e,

$$[\hat{h}_j, \hat{O}] = 0, \quad \forall j$$

Denoting the set of operators $\hat{O}$ as $\mathcal{C}$, we note that operators in $\mathcal{C}$ form a closed associated algebra, i.e.

$$\hat{O}_1 \in \mathcal{C}, \hat{O}_2 \in \mathcal{C} \Rightarrow \begin{cases} \alpha_1 \hat{O}_1 + \alpha_2 \hat{O}_1 \in \mathcal{C} \\ \hat{O}_1 \hat{O}_2 \in \mathcal{C} \end{cases}, \quad \alpha_1, \alpha_2 \in \mathbb{C},$$

hence we refer to $\mathcal{C}$ as the commutant algebra.

If the system contains several types of terms, e.g. $H = \sum_j J_j \hat{h}_j + K_j \hat{g}_j$ with arbitrary coefficients J_j, K_j, we define the commutant as the algebra of operators $\hat{O}$ that individually commute with all types of local terms, i.e.

$$[\hat{h}_j, \hat{O}] = [\hat{g}_j, \hat{O}] = 0, \quad \forall j$$

- An alternative equivalent definition of $\mathcal{C}$ involves the algebra $\mathcal{A}$ generated by arbitrary linear combinations of arbitrary products of the terms of the Hamiltonian $\{\hat{h}_j\}$, and we refer to this as 'bond algebra'

While the identity operator $\mathbb{1}$ is not necessarily generated by these terms, we can always add it to the definition of $\mathcal{A}$ since adding a constant to the family of Hamiltonians does not affect the symmetries.
- $\mathcal{L}(\mathcal{H})$: the algebra of all linear operators on the (finite dimension) Hilbert space, both $\mathcal{C}$ and $\mathcal{A}$ are subalgebras of $\mathcal{L}(\mathcal{H})$.

$\mathcal{L}(\mathcal{H})$: algebra of all linear operators on $\mathcal{H}$

$$\mathcal{C} := \{ \hat{O} \mid \hat{h}_j \hat{O} = \hat{O} \hat{h}_j \ , \ \forall j \}$$

$$\mathcal{A} := \{ \alpha \prod_{i \in I} h_i \mid \alpha \in \mathbb{C} \}$$

in fact this also implies that $\mathcal{A}$ is the centralizer of $\mathcal{C}$

<u>Claim</u> $\mathcal{C}$ is the centralizer of $\mathcal{A}$ in $\mathcal{L}(\mathcal{H})$.

It is easy to see that every elements in $\mathcal{A}$ commutes with $\hat{O}$, $\forall \hat{O}$ #

- Note that $\mathbb{1} \in \mathcal{A}$, $\mathbb{1} \in \mathcal{C}$.
- $\mathcal{A}$ is generated by $\{\hat{h}_j\}$, which are hermitian $\Longrightarrow$ $\mathcal{A}$ is closed under hermitian conjugate.

Suppose $\hat{h}_j \hat{O} = \hat{O} \hat{h}_j$, then $\hat{O}^\dagger \hat{h}_j = \hat{O}^\dagger \hat{h}_j^\dagger = (\hat{h}_j \hat{O})^\dagger = (\hat{O} \hat{h}_j)^\dagger = \hat{h}_j \hat{O}^\dagger$,

hence $\mathcal{C}$ is closed under hermitian conjugate.

$\Longrightarrow$ $\mathcal{A}$, $\mathcal{C}$ are von Neumann algebra.

<u>Claim</u> the centers of $\mathcal{A}$ and $\mathcal{C}$ coincide, given by $Z := \mathcal{A} \cap \mathcal{C}$.

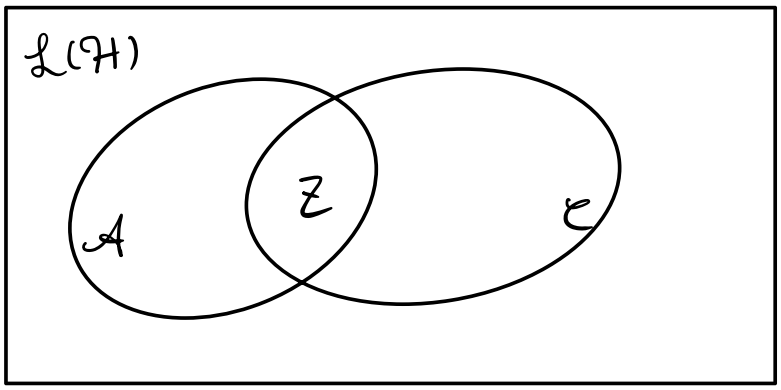
$$\text{Center}(\mathcal{A}) := \{ \ell \in \mathcal{L}(\mathcal{H}) \mid \ell a = a \ell \ , \ a \in \mathcal{A} \}$$

$$\text{Center}(\mathcal{C}) := \{ \ell \in \mathcal{L}(\mathcal{H}) \mid \ell \hat{O} = \hat{O} \ell \ , \ \hat{O} \in \mathcal{C} \}$$

For $\ell \in \text{Center}(\mathcal{A})$, we have $[\ell, a] = 0$, then $[\ell, \hat{O}] = 0$ since $[a, \hat{O}] = 0$. Thus $\ell \in \text{Center}(\mathcal{A})$.

Similarly, $\ell \in \text{Center}(\mathcal{C})$ implies $\ell \in \text{Center}(\mathcal{A})$.

Hence $\text{Center}(\mathcal{A}) = \text{Center}(\mathcal{C})$ #



- We always restrict to systems with tensor product Hilbert spaces with total dimension D.

$$\mathcal{C} \text{ algebra} \longleftrightarrow \text{respective } D\text{-dim representation}$$

$$\text{operator} \longleftrightarrow D\text{-dim matrix}$$

$$\text{orthonormal basis of } \mathcal{C} \longleftarrow \text{orthonormal basis}$$

$$\text{gen}(\mathcal{C}) \overset{\text{in general}}{\neq} \dim(\mathcal{C}) \longleftarrow \text{number of elements in basis}$$

(number of generators)

- The full Hilbert space $\mathcal{H}$ can be decomposed into representations of $\mathcal{A}\times\mathcal{C}$ as follows:

$$\mathcal{H} = \bigoplus_\lambda \left(\mathcal{H}_\lambda^{(\mathcal{A})} \otimes \mathcal{H}_\lambda^{(\mathcal{C})} \right)$$

where $\mathcal{H}_\lambda^{(\mathcal{A})}$ and $\mathcal{H}_\lambda^{(\mathcal{C})}$ denote D_λ and d_λ dimensional irreducible representation of $\mathcal{A}$ and $\mathcal{C}$.

- For each λ, there exists a basis $\{ |u_{\lambda,\alpha}\rangle \otimes |v_{\lambda,\beta}\rangle \}$, $\alpha = 1, \cdots, D_\lambda$, $\beta = 1, \cdots, d_\lambda$,

$$\begin{cases} \hat{h}_{\mathcal{A}} |u_{\lambda,\alpha}\rangle \otimes |v_{\lambda,\beta}\rangle = \sum_{\alpha'} M^\lambda_{\alpha\alpha'}(\hat{h}_{\mathcal{A}}) \, |u_{\lambda,\alpha'}\rangle \otimes |v_{\lambda,\beta}\rangle \\ \hat{h}_{\mathcal{C}} |u_{\lambda,\alpha}\rangle \otimes |v_{\lambda,\beta}\rangle = \sum_{\beta'} N^\lambda_{\beta\beta'}(\hat{h}_{\mathcal{C}}) \, |u_{\lambda,\alpha}\rangle \otimes |v_{\lambda,\beta'}\rangle \end{cases}$$

$\hat{h}_{\mathcal{A}} \in \mathcal{A}$. $\hat{h}_{\mathcal{C}} \in \mathcal{C}$, $\begin{cases} M^\lambda_{\alpha\alpha'}(\hat{h}_{\mathcal{A}}) : D_\lambda \times D_\lambda \text{ matrices} \\ N^\lambda_{\beta\beta'}(\hat{h}_{\mathcal{C}}) : d_\lambda \times d_\lambda \text{ matrices} \end{cases}$

$\Longrightarrow$ operators in $\mathcal{A}$ and $\mathcal{C}$ have the matrix representation:

$$\hat{h}_{\mathcal{A}} = \bigoplus_\lambda \left(M^\lambda(\hat{h}_{\mathcal{A}}) \otimes \mathbb{1}_{d_\lambda} \right)$$

$$\hat{h}_{\mathcal{C}} = \bigoplus_\lambda \left(N^\lambda \mathbb{1}_{D_\lambda} \otimes N^\lambda(\hat{h}_{\mathcal{C}}) \right)$$

- $\dim(\mathcal{A}) = \sum_\lambda D_\lambda$, $\dim(\mathcal{C}) = \sum_\lambda d_\lambda$, $\dim(\mathcal{H}) = \sum_\lambda D_\lambda d_\lambda$

- For $\hat{h}_z \in \mathcal{Z}$, we have

$$\hat{h}_z = \bigoplus_\lambda \left(c_\lambda(\hat{h}_z) \left(\mathbb{1}_{D_\lambda} \otimes \mathbb{1}_{d_\lambda} \right) \right) \quad , \quad c_\lambda(\hat{h}_z) \in \mathbb{C}$$

- Since the time-evolution operator e^{-iHt} is an element of $\mathcal{A}$, it only acts on the basis elements of the first factor $\mathcal{H}_\lambda^{(\mathcal{A})}$ while leaving the second factor $\mathcal{H}_\lambda^{(\mathcal{C})}$ invariant. Hence, for each λ, we obtain d_λ number of D_λ-dimensional subspaces that are invariant under time-evolution.

$$K = \sum_\lambda d_\lambda = \dim(\mathcal{M})$$

↳ number of Krylov subspace

Examples

1. No symmetries

$$\mathcal{C} = \{ \mathbb{1} \} \quad , \quad \mathcal{A} = \mathcal{L}(\mathbb{H})$$

↳ $\dim(\mathcal{C}) = \mathrm{gen}(\mathcal{C}) = 1$

$$\hookrightarrow \sum_\lambda d_\lambda^2 = 1 \implies d_\lambda = 1 \implies D_\lambda = \dim(\mathbb{H})$$

$$\dim(\mathcal{A}) = |\dim(\mathbb{H})|^2$$

$$K = \sum_\lambda d_\lambda = 1 \implies \text{single dynamically disconnected Krylov subspace}$$

2. Abelian symmetries

$$H_{XXZ} = \sum_{j=1}^{L} \left[J_j^\perp \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y \right) + J_j^z S_j^z S_{j+1}^z \right] + \sum_{j=1}^{L} h_j S_j^z$$

conserved quantity: $S_{tot}^z = \sum_{j=1}^{L} S_j^z$

S^z_{tot} is part of $\mathcal{C}$ corresponding to the bond algebra $\mathcal{A}$ generated by $\{ (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y) \}$ and $\{ S_j^z \}$

$\mathcal{C}$ is precisely spanned by all powers of S^z_{tot} along with $\mathbb{1}$.

$*$ $(S_j^z)^2 = \frac{\mathbb{1}}{4}$, $\Rightarrow (S^z_{tot})^{L+1}$ can be expressed in terms of lower powers of S^z_{tot}

$$\Rightarrow \mathcal{C} = \text{span}\{ \mathbb{1}, S^z_{tot}, \ldots, (S^z_{tot})^L \}$$

$\dim(\mathcal{C}) = L+1$

$d_\lambda = 1$ for all λ $\Rightarrow$ λ runs over $(L+1)$ values

$\Updownarrow$

total spin for a spin $\frac{1}{2}$ system with L spins can take

$$\underbrace{-\frac{L}{2}, -\frac{L+1}{2}, \ldots, \frac{L-1}{2}, \frac{L}{2}}_{L+1}$$

3. Non-Abelian symmetries

$$H_{Heis} = \sum_j J_j \vec{S}_j \cdot \vec{S}_{j+1}$$

$SU(2)$ symmetry. $\quad [\vec{S}_j \cdot \vec{S}_{j+1}, \sum_{i=1}^{L} S_i^\alpha] = 0$

$1 \le j \le L, \quad \alpha = x, y, z$

The full $\mathcal{C}$ is the associative algebra consisting of all products and linear combinations of S^x_{tot}, S^y_{tot}, S^z_{tot}

? ? ?

4. Solvable models

$D_\lambda = 1$ for all λ

$K = \dim(\mathcal{H}) \Rightarrow$ All eigenstates are one-dim Krylov space.

solvable

$\dim \mathcal{A} = \sum_\lambda D_\lambda^2$

$\dim \mathcal{C} = \sum_\lambda d_\lambda^2$

$\dim \mathcal{H} = \sum_\lambda D_\lambda d_\lambda$

$K = \sum_\lambda d_\lambda = \dim(\mathcal{M})$

Hilbert space fragmentation

$\sqrt{\dim \mathcal{C}} \leq K \leq \dim(\mathcal{C})$

① $\log(\dim \mathcal{C})$ independent of L

discrete sym : $\mathbb{Z}_2$

② $\log(\dim \mathcal{C}) \propto \log L$

conti. sym : $U(1)$, $SU(2)$

③ $\log(\dim \mathcal{C}) \propto L$

$\sqrt{e^L} \leq K \leq e^L \rightsquigarrow$ H.S.F

还希望 $\mathcal{A}$ 是 non-abelian, 否则就是 solvable model, i.e. trivial HSF

t-J_z Model

(A). Def. sym.

$$H_{t\text{-}J_z} = \sum_{\langle i,j\rangle} \Big[\overbrace{-t_{i,j} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left(\tilde{c}_{i,\sigma}\tilde{c}^{\dagger}_{j,\sigma} + H.c.\right)}^{\hat{T}_{i,j}} + \overbrace{J^z_{i,j} S^z_i S^z_j}^{\hat{V}_{i,j}} \Big]$$

$$+ \underbrace{\sum_i \left[h_i S^z_i + g_i (S^z_i)^2 \right]}_{\text{破坏 discrete sym.}}$$

两个 $U(1)$ sym: $N^{\uparrow}$, $N^{\downarrow}$, $N^{\sigma}_j := \tilde{c}^{\dagger}_{j,\sigma}\tilde{c}_{j,\sigma}$

(B). Fragmentation

$\hat{T}_{i,j}$: $|\uparrow 0\rangle \leftrightarrow |0\uparrow\rangle$

$|\downarrow 0\rangle \leftrightarrow |0\downarrow\rangle$

$\Longrightarrow$ $|\uparrow\downarrow 0\downarrow\uparrow\rangle$ 和 $|\downarrow\uparrow\downarrow 0\uparrow\rangle$ 动力学不连通

即使它们的 $N^{\uparrow}, N^{\downarrow}$ 相同 ($\rightsquigarrow$ HSF)

OBC: $\sum_{N=0}^{L} 2^N = 2^{L+1} - 1$ 个子空间

PBC: 懂的都懂 :)

$\rightarrow$ $N=L$ 的时候还有指数多的 frozen states, 在它们上面 $\hat{T}_{i,j}$ vanishes.

(C). Commutant Algebra

$[N^{\sigma}_i N^{\tau}_j, \hat{T}_{i,j}] = 0$ $\quad \sigma,\tau \in \{\uparrow,\downarrow\}$

$[N^{\sigma}_i N^{\tau}_j, \hat{V}_{i,j}] = 0$

$$N^{\sigma_1\sigma_2} := \sum_{j_1<j_2} N_{j_1}^{\sigma_1} N_{j_2}^{\sigma_2}, \quad \sigma_1,\sigma_2 \in \{\uparrow,\downarrow\}$$

$$N^{\sigma_1\cdots\sigma_k} = \sum_{j_1<j_2<\cdots<j_k} N_{j_1}^{\sigma_1} N_{j_2}^{\sigma_2} \cdots N_{j_k}^{\sigma_k} \quad \sigma_j \in \{\uparrow,\downarrow\}$$

↳ 全部线性无关，张成了 $\mathcal{C}_{OBC}$ 的一个完备基

$$\dim(\mathcal{C}_{OBC}) = \sum_{k=0}^{L} 2^k = 2^{L+1} - 1$$

Pair-Flip Model　　spin-$\frac{m-1}{2}$ (m-level)

(A) Def. sym.

$$H_{PF}^{(m)} := \sum_{\langle i,j\rangle} \sum_{\alpha,\beta=1}^{m} \overbrace{\left[g_{i,j}^{\alpha,\beta} \left(|\alpha\alpha\rangle\langle\beta\beta| \right)_{i,j} + H.c. \right]}^{\hat{F}_{i,j}^{\alpha,\beta}}$$

$$+ \underbrace{\sum_i \sum_{\alpha=1}^{m} \lambda_i^{\alpha} \underbrace{\left(|\alpha\rangle\langle\alpha| \right)_i}_{N_i^{\alpha}}}_{\text{被不 discrete sym.}}$$

$$U(1) : \quad N^{\alpha} := \sum_j (-1)^j N_j^{\alpha} \qquad 1 \le \alpha \le m$$

$$[N_i^{\alpha} - N_j^{\alpha}, \hat{F}_{i,j}^{\beta,\gamma}] = 0 \qquad * \sum_{\alpha=1}^{m} N_j^{\alpha} = \mathbb{1}$$

$$[N_i^{\alpha}, N_i^{\beta}] = 0$$

$\Longrightarrow$ PF model has a $U(1)^{m-1}$ sym.

Temperley - Lieb Models

(A) Def. sym.

$$H_{TL}^{(m)} = \sum_{j=1}^{L-1} J_j \hat{e}_{j,j+1} := \sum_{j=1}^{L-1} J_j \left[\sum_{\alpha,\beta=1}^{m} (|\alpha\alpha\rangle\langle\beta\beta|)_{j,j+1} \right]$$